



IT Employee Lifecycle Automation System

Automate onboarding and offboarding — zero manual steps, zero orphaned accounts.

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The Problem: Manual IT Account Management

When a new employee joins:

- IT receives notification via email or ticket — often the day of or after the start date
- Admin must manually create an AD account, assign security groups, and verify access
- Result: **delayed first-day access, lost productivity**

When an employee leaves:

- IT frequently receives **late or incomplete notifications**
- Former employee retains active credentials for days or weeks
- **Orphaned accounts** are a common vector for data breaches

Additional Issues:

- Different IT staff assign different permissions to the same role → **access disparities**
- No centralized audit trail → **compliance failures**
- Manual effort is time-consuming, error-prone, and hard to scale

The Solution: Automated Lifecycle Management

What the system does:

```
OrangeHRM detects change
|
v
Backend poller fires automatically
|
+-- New hire?    → Create AD account + assign groups → Notify IT
|
+-- Termination? → Disable AD account + remove groups → Notify IT
```

No tickets. No emails. No manual steps.

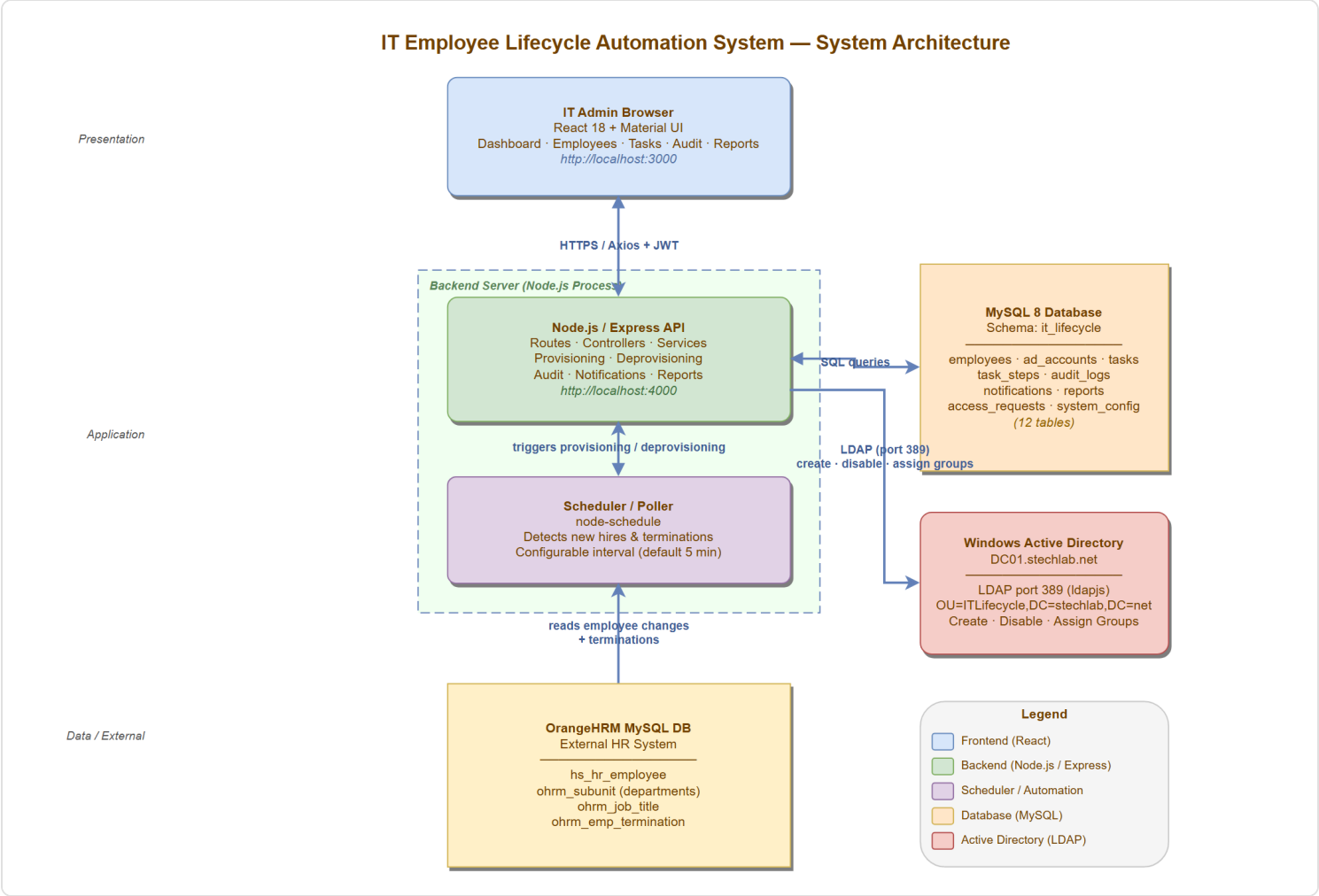
IT administrators retain a full dashboard for oversight and manual override.

Every action is logged for compliance review.

Key Benefits:

- New hires have accounts ready on **Day 1**
- Terminated employees are disabled within **minutes**
- Consistent permissions via **role-based group mappings**
- Complete **audit trail** for compliance and security reviews
- IT team freed from repetitive provisioning tasks

System Architecture



Stack: React · Node.js · Express · MySQL · Idapjs · JWT · node-schedule

What You'll See in the Demo

#	Feature	What it demonstrates
1	Dashboard	Real-time task stats + manual OrangeHRM sync
2	Employee Search	Find employee, click to view AD account detail
3	Task Management	Filter tasks, flag as Critical, view retry logic
4	Audit Log	Date-range filtering + CSV export
5	Access Requests	Submit and approve additional-access requests
6	Reports	Generate compliance report + download CSV
7	System Config	Adjust polling interval, retry limit, notifications

Total runtime: ~7 minutes

10 Implemented Features

Feature	Requirement(s)
OrangeHRM DB polling (name, dept, title sync)	REQ-1, REQ-4, NFR-1
Automatic AD account provisioning via LDAP	REQ-2, NFR-2
Role-based AD group assignment from mappings	REQ-3, REQ-6
Automatic deprovisioning on termination	REQ-5
Task management with retry logic + priority flag	REQ-7, REQ-12, NFR-5
Audit log with date filter and CSV export	REQ-9
In-app notifications with unread badge	REQ-15
Access request workflow (submit / approve / reject)	REQ-14
Compliance report generation + CSV download	REQ-10
System configuration UI (no code changes needed)	REQ-8, NFR-10

All 15 functional requirements implemented.

Live Demo

Switch to browser — <http://localhost:3000>

Demo Stops:

1. **Dashboard** → stat cards + Sync Now button
2. **Employee Search** → click row → AD account modal
3. **Task List** → filters, priority badge, flag Critical
4. **Audit Log** → set date range → Export CSV
5. **Access Requests** → approve a request
6. **Reports** → generate + download CSV
7. **Settings** → polling interval, retry limit, notifications toggle

Progress vs. Original Plan

Week	Planned	Actual
1-4	Project proposal + system requirements	Complete
5-7	Environment setup + core dashboard	Complete
8	Mid-term: core onboarding automation	Complete
9-11	Offboarding, notifications, audit log	Complete
12-14	Reports, access requests, config UI	Complete

Ahead of schedule: All 15 requirements done at Week 8 milestone. Full feature set (10 features) implemented and running end-to-end on live AD + OrangeHRM.

Remaining Work (Weeks 12–15)

Quality & Security

- LDAPS (encrypted LDAP on port 636) — currently plain LDAP port 389
- Input validation hardening on all API endpoints

Role-Based UI

- HR Personnel view: read-only employee status, flag high-priority tasks
- Department Manager view: team account status + access request submission

Testing

- Unit tests (Jest): provisioningService, deprovisioningService, ADConnector, notificationService
- Integration tests: end-to-end lifecycle flows against a test DB instance

Documentation

- Final system documentation with test results
- README updates for deployment

Summary

Problem solved: Manual IT provisioning causes delayed access, orphaned accounts, and compliance risk.

Solution delivered: Fully automated lifecycle system that detects HR changes and acts on them — no human in the loop required for standard cases.

Technical Achievement:

- Live integration: OrangeHRM MySQL → Node.js poller → LDAP → Active Directory
- 12-table MySQL schema, 15 API routes, 9 React pages
- Retry logic, priority queuing, full audit trail, CSV exports

Still to Come:

- LDAPS, role-scoped UI views, automated test suite

Thank you — Questions?